

PALLETI VIRAJA

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EDUCATION

Shiv Nadar University, Delhi NCR

Aug 2024 – Present

B.Tech in Computer Science and Engineering, Second Year, GPA: 7.8

Narayana Junior College, Hyderabad

May 2024

12th Grade: 95.2%

JEE Mains: 96.5 Percentile

EXPERIENCE

Research Intern – IIIT Hyderabad – Language Technologies Research Centre (LTRC)

May – Aug 2026

Advisor: Prof. Chiranjeevi Yarra

- Developing an automatic prosodic phrase classifier for English intonational categories, with a custom-annotated ToBI dataset and full speech processing pipeline.

PROJECTS & PAPERS

Hermes: Low-Communication Distributed LLM Training — *[Github link: Paper (in submission)]*

- Designed a distributed training system with GPU-level parallelism via custom CUDA kernels, cutting gradient-compression latency from 160ms to 3.5ms (46×) and communication bandwidth up to 13,000× versus standard distributed data parallel (DDP).
- Introduces heterogeneity-aware sharding enabling LLM training across mixed hardware over commodity WiFi.

BlindSpot: Hardware Navigation System for the Visually Impaired — *[GitHub link]*

- Won 1st place among 40+ teams; deployed YOLOv8 on Raspberry Pi using NCNN for ARM-optimised real-time inference, with directional haptic feedback via a companion mobile app.
- *Stack:* Python, YOLOv8, NCNN, Raspberry Pi, Flask, Svelte, OpenCV

Refract: Style-Adaptive PDF Assistant — *[Github link: in progress]*

- An intelligent document assistant that ingests PDFs and dynamically adapts explanations to the user's own writing style using transformer-based semantic similarity.
- *Stack:* Python, PyPDF2, HuggingFace Transformers, Sentence Transformers, pyttax3

Dungeon Master: Interactive RPG — *[GitHub link]*

- Stateful RPG engine managing persistent game state across turns using LangGraph, with tool-calling for game mechanics and human-in-the-loop interrupts for turn-based gameplay.
- *Stack:* Python, LangGraph, LLM Prompt Engineering

PageIt: Offline Encyclopedia — *[GitHub link]*

- Fine-tuned Gemma-2B on-device for offline factual Q&A on Android, doubling as an SMS server to distribute articles to users in low-connectivity regions.
- *Stack:* Kotlin, Python, HuggingFace

LEADERSHIP & ACTIVITIES

- **Winner** – Hackdata 2026 (Hackathon)
- **Secretary of the Treasury** – Shiv Nadar University Student Council 25–26
- Top 10 teams (SNU) – Smart India Hackathon 2025
- ACM SNIOE: Data Analytics team member, Marketing member
- Astronautics & Aeronautics Society, SNIOE: Technical team
- Google Developers Group, SNIOE: AI/ML team member

SKILLS & INTERESTS

Languages: C++, Python, C, Java, JavaScript

Systems & Infrastructure: SQL & NoSQL databases, AWS, CUDA, Git, AZ-900 (in progress)

AI/ML: PyTorch, Scikit-learn, HuggingFace, LangChain, LangGraph

Web/Backend: Flask, React, Node

Tools: Linux, Raspberry Pi

Relevant Coursework: Data Structures, Computer Architecture, Object-Oriented Programming, Introduction to LLMs, Introduction to Machine Learning